

Explicit sequences



- 1 a Finite b Infinite
- 2 6
- 3 196
- 4 a $\frac{5}{4}, \frac{6}{5}, \frac{7}{6}, \frac{8}{7}, \frac{9}{8}$ b $\frac{2}{3}, 1, 1, \frac{19}{21}, \frac{4}{5}$
c $-1, -\frac{1}{4}, 0, \frac{1}{16}, \frac{1}{16}$ d 0, 3, 6, 9, 12
e $-2, -4, -8, -16, -32$ f $-6, 24, -72, 192, -480$
g $1, 0.7071, 0.5774, 0.5, 0.4472$ h $2, 2.25, 2.3704, 2.4414, 2.4883$
- 5 $u_n = 3 \times 0.1^{n-1}$
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Applications

- 6 a \$1, \$3, \$7, \$15 b \$38, \$43, \$48, \$53
c Day 6
- 7 a $u_n = 2n + 7$ b 27
- 8 a $u_n = 8.25 (0.55)^{n-1}$ b 0.75 metres